

BlindVision — Supplementary Material

Supplementary Tables S1–S18 accompany the main manuscript. They contain protocol, dataset, failure-mode, power, reliability, software-environment, computational, comparison, navigation, ablation, sensor, latency, subsystem-result, detection-metric, and reconciliation details moved out of the main manuscript to improve readability.

TABLE S1. Dataset composition.

Split	Images	Annotated objects
Train	5,600	— (not reported per-split)
Validation	700	— (not reported per-split)
Test (held-out)	700	842

TABLE S2. Detection success by lighting and occlusion.

Condition	Detection success rate
Bright daylight	97.8%
Low light	87.9%
No occlusion	97.5%
Severe occlusion (>75%)	74.2%

TABLE S3. Experimental protocol summary.

Item	Value
Evaluation sessions (full prototype)	20
Total navigation trials	100
Approx. total walking distance	~2 km
Object detection attempts	200
BLE packets transmitted (test)	8,000
Visually impaired (BVI) participants, total	7 (1 earlier + 6 later)
Additional BVI-specific trials (six-participant session)	48 (8 per participant: 3 outdoor nav, 3 stair/drop-off, 2 water hazard)
Sighted testers	Project-team members (count not specified)

TABLE S4. Power budget.

System	Approx. input power / basis
Smart Goggles (5-V USB input)	12.11 W mean, measured from five released active-detection samples
Smart Stick (3.7-V nominal cell)	≈0.8 W, derived/assumed for runtime projection
Combined	Not directly measured; separate power domains

TABLE S5. Battery/runtime characterization.

System	Measured / derived runtime	Basis
Goggles — idle	Not reported	No idle samples in released power log
Goggles — active detection	≈3.1 h nominal-equivalent	37 Wh nominal / 12.11 W measured mean; projection, not depletion test
Goggles — maximum simultaneous load	Not reported	No maximum-load samples in released power log
Smart Stick	≈8.7 h upper bound	Derived from 5,000 mAh nominal capacity at 575 mA assumed average draw; no continuous discharge test

TABLE S6. Reliability (field measurements).

Metric	Result
Continuous operating duration tested	4 hours
Crashes observed	0
Restarts observed	0
Sustained RAM growth observed	0 (none)

TABLE S7. Software environment.

Component	Version / detail
On-device detection model	YOLOv8n (nano), CPU-only, no hardware accelerator
Host (full prototype)	Raspberry Pi 4 Model B, 4 GB
Operating system	Raspberry Pi OS (exact build/architecture not specified in the release)
ultralytics (Smart Goggles)	>=8.2.0
ultralytics (proof-of-concept cloud server)	==8.2.28
fastapi	==0.111.0
uvicorn[standard]	==0.30.1
pydantic	==2.7.4
opencv-python	>=4.9.0
bleak (BLE)	>=0.22.0
pyttsx3 (offline TTS)	>=2.90
pyserial	>=3.5
pynmea2 (GPS parsing)	>=1.19.0
numpy	>=1.26.0
pillow (proof-of-concept)	==10.3.0
python-multipart (proof-of-concept)	==0.0.9

TABLE S8. Ten-class confusion matrix on the 700-image held-out test split.

True \ Predicted	Person	Door	Chair	Backpack	Laptop	Bottle	Pole	Vehicle	Bicycle	Stairs
Person	110	0	1	0	0	0	1	2	0	1
Door	1	75	0	0	1	0	0	0	1	0
Chair	0	0	88	1	0	0	0	0	0	3
Backpack	0	1	0	61	0	1	0	0	1	1
Laptop	0	0	0	0	53	1	0	0	0	1
Bottle	0	0	0	0	0	66	1	0	0	3
Pole	1	0	0	1	0	0	120	0	0	2
Vehicle	1	1	0	0	0	0	0	84	1	1
Bicycle	0	0	1	1	0	0	0	0	57	1
Stairs	0	0	1	0	0	2	0	1	0	91

TABLE S9. Computational-load measurements from the released run.

Metric	Condition	Measured value
CPU utilization (avg., 4 cores)	During active detection + fusion	61% (psutil)
CPU utilization (peak)	During active detection + fusion	87% (psutil)
RAM usage	During active detection + fusion	1.7 GB (psutil)
Processor temperature	During active detection + fusion	61 °C (vcgencmd)
Storage utilization	64 GB microSDXC (OS, models, logs)	12.3 GB occupied / 51.7 GB free
Camera pipeline frame rate (FPS)	Four concurrent streams, 60 s run (Section VI.I)	22.8 FPS aggregate; 5.7 FPS per-camera mean (measured)
Detection stage speed (single stream)	Per frame, on-device (Raspberry Pi 4)	122 ms/frame (from Table S15)
Detection stage speed (four concurrent streams)	Per camera, four streams active	175 ms/frame (derived as 1/5.7 FPS)
Parallel efficiency, four streams	22.8 FPS measured vs 32.8 FPS if perfectly parallel	69.5% (derived)

TABLE S10. Contextual comparison with related assistive-navigation systems.

System	Vision	Ultrasonic /ground	Offline fallback	Caregiver layer	Latency	End-to-End Detection/
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						Guidance Success
ALVU [2]	No (ToF array, not camera-based)	Yes (waist-level ToF)	N/R	No	N/R	Qualitative , 162 trials
Gao et al. [3]	Yes (RGB+D glasses)	No	No (phone-dependent)	No	<320 ms	100% avoidance
7Sense SuperBrain 1 [6]	Yes (RGB/depth/thermal)	No	N/R	No	N/R	N/R (commercial; no published trial data)
Mai et al. [10]	Yes (RGB-D camera)	Yes (2D LiDAR on cane, not ultrasonic)	N/R (single integrated device, not two independently-operable nodes)	No	N/R	61.2% overall recognition rate across 86 object classes, 25–26 FPS (as reported in [10]; not a directly comparable protocol)
BlindVision	Yes (4-camera YOLOv8)	Yes (5x ultrasonic + 2x IR)	Yes (Vision-Only/Stick-Only modes, Section III)	Backend/API event layer; no released smartphone app	205 ms serialized stage-sum	96.0% end-to-end detection & guidance success; 96.0% fused navigation

TABLE S11. Navigation trial results by scenario.

Scenario	Trials	Successful	95% CI
Corridor navigation	30	29	83.3–99.4%
Outdoor walk	25	24	80.5–99.3%
Stair avoidance	20	19	76.4–99.1%
Water hazard	10	10	72.2–100%
Moving person detection	15	14	70.2–98.8%

TABLE S12. Subsystem ablation and power-basis status.

System	Approx. input power / basis	Successful	Navigation success rate	Navigation success rate
Smart Goggles (5-V USB input)	12.11 W mean, measured from five released active-detection samples	Not reported	Not reported	
Smart Stick (3.7-V nominal cell)	≈0.8 W, derived/assumed for runtime projection	Not reported	Not reported	
Combined	Not directly measured; separate power domains	96	96.0% (overall navigation result)	

TABLE S13. Sensor reading accuracy, full system.

Sensor	Tested range	Mean absolute error	SD / basis
Ultrasonic (historical aggregate)	20–300 cm	2.4 cm	0.7 cm (n=10/point)
IR sensors	5–80 cm	1.8 cm	0.5 cm
Water sensor	Wet/dry	98.4% accuracy (reported aggregate)	n/a (binary)
IMU tilt	±90°	±2°	0.6°
FSR	Light/medium/heavy pressure	48/50 correct (96%)	95% CI 86.5–98.9%

TABLE S14. Reported aggregate water- and stair-hazard detection metrics.

Metric	Value
Water Detection – Accuracy	98.4%
Water Detection – Precision	98.1%
Water Detection – Recall	98.8%
Stair Detection – Precision	96.2%
Stair Detection – Recall	95.5%
Stair Detection – F1	95.8%

TABLE S15. Single-stream processing-path stage breakdown.

Process stage / characterization	Time (mean)	Measurement basis / dispersion
Detection (image capture + YOLO inference)	122 ms	N/A (not reported)
BLE transmission	17.5 ms	N/A (not reported)
Sensor fusion	9 ms	N/A (not reported)
Serialized stage-sum (not	205 ms	Derived from stage means

direct end-to-end)		
Total alert time	205 ms	Derived from stage means

TABLE S16. Full-system experimental results by subsystem, with 95% Wilson score confidence intervals.

Test	Total	Success	Rate	95% CI
Object detection	200	192	96.0%	92.3–98.0%
Voice guidance	200	194	97.0%	93.6–98.6%
BLE communication	8000	7920	99.0%	98.8–99.2%
Ultrasonic detection	180	174	96.7%	92.9–98.5%
SOS alert	20	20	100%	83.9–100%

TABLE S17. Detection results and reconciled metrics for the 700-image held-out split. Precision and F1 are reconstructed from aggregate confusion/false-positive counts; mAP values remain reported outputs because raw prediction-level outputs are unavailable.

Metric	Result
Precision	92.2% (reconciled)
Recall	95.6%
F1-score	93.9% (reconciled)
mAP@0.50	96.8%
mAP@0.50:0.95	76.5%
IoU threshold	0.50
Average confidence threshold	0.45
Test images	700
Test object classes	10
Capture-to-alert latency	~0.7–0.9 s (midpoint ~0.8 s)

TABLE S18. Reconciled detection false-positive accounting.

Metric	Value
Sensitivity	95.6%
False-alarm rate (1 – precision)	7.8% (from reconciled precision)
False Negative Rate	4.4%
False positives per image	0.097 (68 / 700)

ADDITIONAL EXPERIMENTAL RECORDS

The following tables document the additional author-supplied experiments added after the earlier release. Raw row-level records are retained in data/raw/ and are not replaced by these summaries.

TABLE S19. Recomputed per-transducer calibration.

Sensor	m	b	R ²	Corrected MAE (cm)	Corrected SD (cm)	Max error (cm)
U1	0.99869704	0.51718625	0.999894	0.707	0.940	1.976
U2	1.00029776	-0.01266986	0.999968	0.431	0.514	0.923
U3	0.99884492	0.30787370	0.999931	0.583	0.759	1.540
U4	0.99796163	0.40065947	0.999944	0.552	0.684	1.294
U5	0.99962230	0.22446043	0.999984	0.303	0.367	0.689

TABLE S20. Incidence-angle characterization at 100 cm.

Angle (deg)	n	Mean measured (cm)	Mean error (cm)	Mean relative amplitude
-60	5	105.44	5.44	0.346
-45	5	102.14	2.14	0.646
-30	5	101.22	1.22	0.846
-15	5	100.54	0.54	0.952
0	5	100.30	0.30	1.000
15	5	100.64	0.64	0.960
30	5	101.40	1.40	0.836
45	5	102.82	2.82	0.620
60	5	106.14	6.14	0.320

TABLE S21. Target-material characterization at 100 cm.

Material	n	Mean measured (cm)	Mean error (cm)	SD (cm)
flat_board	5	101.94	1.94	0.34
concrete	5	101.42	1.42	0.26
metal	5	100.44	0.44	0.23
fabric	5	109.12	9.12	0.61

TABLE S22. Simultaneous-firing cross-talk check.

Sensor	Solo mean at 100 cm	Simultaneous reading	Shift (cm)
U1	100.70	101.50	0.80
U2	100.20	102.10	1.90
U3	100.50	101.80	1.30
U4	100.70	102.40	1.70
U5	100.55	101.90	1.35

TABLE S23. Fusion-weight sensitivity summary.

Configuration	Changed decisions (of 10)	Interpretation
Baseline	0	Reference configuration
C +20%	1	One decision changed
C -20%	0	All preserved
P +20%	0	All preserved
P -20%	1	One decision changed
U +20%	0	All preserved
U -20%	0	All preserved
W +20%	0	All preserved
W -20%	0	All preserved

TABLE S24. Trial-level paired subsystem ablation.

Condition	Trials	Successes	Collisions	Missed hazards	Mean completion time (s)
full	4	4	0	0	53.58
no_vision	4	0	4	0	35.90
no_ultrasonic	4	2	2	3	58.23
no_fusion	4	0	4	3	43.60

TABLE S25. Continuous Smart Stick discharge run.

Elapsed (min)	Voltage (V)	Current (A)	State
0	4.2	0.42	active
30	4.05	0.41	active
60	3.91	0.4	active
90	3.78	0.4	active
120	3.65	0.39	active
150	3.52	0.39	active
180	3.38	0.38	active
210	3.2	0.37	active
240	3.01	0.35	cutoff_reached

TABLE S26. Raw experimental records included in the repository.

Raw file	Rows	Purpose
data/raw/ Ultrasonic_Calibration_Raw.csv	80	Per-transducer distance calibration
data/raw/ Ultrasonic_Calibration_With_Correction_Raw.csv	80	Raw readings plus inverse-corrected values
data/raw/ Ultrasonic_Angle_Sweep_Raw.csv	45	Incidence-angle response, five sensors $\times$ nine angles
data/raw/ Ultrasonic_Material_Raw.csv	20	Five sensors $\times$ four target materials $\times$ one supplied observation per sensor/material at 100 cm
data/raw/ Ultrasonic_Crosstalk_Raw.csv	5	One simultaneous-firing observation per sensor

data/raw/ Fusion_Weight_Sensitivity_10 cases.csv	10	Inputs for ten recorded fusion cases
data/raw/ Fusion_Weight_Sensitivity_Su mmmary.csv	9	Nine tested weight configurations
data/raw/ Paired_Subsystem_Ablation_ Raw.csv	16	Four paired navigation trials × four conditions
data/raw/ Smart_Stick_Continuous_Disc harge_Raw.csv	9	Single 240-min discharge record

APPENDIX A. RAW RECORDS

The following tables reproduce the newly added raw author-supplied records row-for-row. They are included here in addition to the CSV files in the repository; no earlier aggregate records are replaced or deleted.

Ultrasonic_Calibration_Raw.csv

sensor_id	true_distance_cm	measured_distance_cm	trial
U1	40	40.5	1
U1	40	40.8	2
U1	60	60.2	1
U1	60	60.6	2
U1	80	79.9	1
U1	80	80.4	2
U1	100	100.3	1
U1	100	101.1	2
U1	150	149.5	1
U1	150	150.8	2
U1	200	199.2	1
U1	200	201.0	2
U1	250	248.9	1
U1	250	251.4	2
U1	300	298.5	1
U1	300	302.1	2
U2	40	39.8	1
U2	40	40.3	2
U2	60	59.7	1
U2	60	60.4	2
U2	80	80.1	1
U2	80	79.6	2
U2	100	99.9	1
U2	100	100.5	2
U2	150	150.2	1
U2	150	149.7	2
U2	200	200.4	1
U2	200	199.5	2
U2	250	249.2	1
U2	250	250.8	2
U2	300	301.0	1
U2	300	299.4	2
U3	40	40.1	1
U3	40	39.9	2
U3	60	60.5	1
U3	60	59.8	2
U3	80	80.2	1
U3	80	80.7	2
U3	100	100.1	1
U3	100	100.9	2
U3	150	149.8	1
U3	150	150.4	2
U3	200	201.2	1
U3	200	198.9	2
U3	250	250.3	1
U3	250	249.1	2
U3	300	298.8	1
U3	300	301.5	2
U4	40	40.4	1
U4	40	39.6	2
U4	60	60.1	1
U4	60	60.7	2
U4	80	79.8	1
U4	80	80.3	2
U4	100	100.4	1
U4	100	101.0	2
U4	150	150.6	1
U4	150	149.2	2
U4	200	199.8	1
U4	200	200.7	2
U4	250	251.1	1
U4	250	248.6	2
U4	300	300.2	1
U4	300	299.1	2
U5	40	39.9	1
U5	40	40.2	2
U5	60	60.3	1

U5	60	59.9	2
U5	80	80.0	1
U5	80	80.5	2
U5	100	100.3	1
U5	100	100.8	2
U5	150	149.9	1
U5	150	150.5	2
U5	200	200.1	1
U5	200	199.8	2
U5	250	249.7	1
U5	250	250.4	2
U5	300	299.6	1
U5	300	300.8	2

Ultrasonic_Angle_Sweep_Raw.csv

sensor_id	angle_deg	true_distance_cm	measured_distance_cm	relative_amplitude	signal_valid
U1	-60	100	105.4	0.35	1
U1	-45	100	102.1	0.65	1
U1	-30	100	101.2	0.85	1
U1	-15	100	100.5	0.95	1
U1	0	100	100.3	1.00	1
U1	15	100	100.6	0.96	1
U1	30	100	101.4	0.84	1
U1	45	100	102.8	0.62	1
U1	60	100	106.1	0.32	1
U2	-60	100	105.8	0.33	1
U2	-45	100	102.5	0.63	1
U2	-30	100	101.5	0.83	1
U2	-15	100	100.7	0.94	1
U2	0	100	100.4	1.00	1
U2	15	100	100.8	0.95	1
U2	30	100	101.6	0.82	1
U2	45	100	103.1	0.60	1
U2	60	100	106.5	0.30	1
U3	-60	100	105.2	0.36	1
U3	-45	100	101.9	0.66	1
U3	-30	100	101.0	0.86	1
U3	-15	100	100.4	0.96	1
U3	0	100	100.1	1.00	1
U3	15	100	100.5	0.97	1
U3	30	100	101.2	0.85	1
U3	45	100	102.6	0.64	1
U3	60	100	105.9	0.34	1
U4	-60	100	105.5	0.34	1
U4	-45	100	102.2	0.64	1
U4	-30	100	101.3	0.84	1
U4	-15	100	100.6	0.95	1
U4	0	100	100.4	1.00	1
U4	15	100	100.7	0.96	1
U4	30	100	101.5	0.83	1
U4	45	100	102.9	0.61	1
U4	60	100	106.2	0.31	1
U5	-60	100	105.3	0.35	1
U5	-45	100	102.0	0.65	1
U5	-30	100	101.1	0.85	1
U5	-15	100	100.5	0.96	1
U5	0	100	100.3	1.00	1
U5	15	100	100.6	0.96	1
U5	30	100	101.3	0.84	1
U5	45	100	102.7	0.63	1
U5	60	100	106.0	0.33	1

Ultrasonic_Material_Raw.csv

sensor_id	target_material	true_distance_cm	measured_distance_cm	incidence_angle_deg	trial	signal_valid
U1	flat_board	100	101.5	0	1	1
U2	flat_board	100	102.1	0	1	1
U3	flat_board	100	101.8	0	1	1
U4	flat_board	100	102.4	0	1	1
U5	flat_board	100	101.9	0	1	1
U1	concrete	100	101.1	0	1	1
U2	concrete	100	101.5	0	1	1
U3	concrete	100	101.3	0	1	1
U4	concrete	100	101.8	0	1	1
U5	concrete	100	101.4	0	1	1
U1	metal	100	100.2	0	1	1
U2	metal	100	100.5	0	1	1
U3	metal	100	100.3	0	1	1

U4	metal	100	100.8	0	1	1
U5	metal	100	100.4	0	1	1
U1	fabric	100	108.5	0	1	1
U2	fabric	100	109.2	0	1	1
U3	fabric	100	108.8	0	1	1
U4	fabric	100	110.1	0	1	1
U5	fabric	100	109.0	0	1	1

Ultrasonic_Crosstalk_Raw.csv

sensor_id	true_distance_cm	mode	target_material	incidence_angle_deg	trial	measured_distance_cm	valid
U1	100	simultaneous	flat_board	0	1	101.5	1
U2	100	simultaneous	flat_board	0	1	102.1	1
U3	100	simultaneous	flat_board	0	1	101.8	1
U4	100	simultaneous	flat_board	0	1	102.4	1
U5	100	simultaneous	flat_board	0	1	101.9	1

Fusion_Weight_Sensitivity_10cases.csv

case_id	C	P	U	W
1	0.85	0	0.9	0
2	0.62	0	0.7	1
3	0.91	0	0.85	0
4	0.55	0	0.6	0
5	0.78	0	0.95	1
6	0.45	0	0.5	0
7	0.88	0	0.8	1
8	0.72	0	0.65	0
9	0.95	0	0.9	1
10	0.58	0	0.55	0

Paired_Subsystem_Ablation_Raw.csv

trial_id	route	condition	success	collision	missed_hazard	completion_time_s
1	R1	full	1	0	0	45.2
1	R1	no_vision	0	1	0	32.5
1	R1	no_ultrasonic	1	0	1	50.1
1	R1	no_fusion	0	1	1	38.4
2	R1	full	1	0	0	44.8
2	R1	no_vision	0	1	0	30.1
2	R1	no_ultrasonic	0	1	1	48.6
2	R1	no_fusion	0	1	0	35.2
3	R2	full	1	0	0	62.4
3	R2	no_vision	0	1	0	41.2
3	R2	no_ultrasonic	1	0	1	68.9
3	R2	no_fusion	0	1	1	51.0
4	R2	full	1	0	0	61.9
4	R2	no_vision	0	1	0	39.8
4	R2	no_ultrasonic	0	1	0	65.3
4	R2	no_fusion	0	1	1	49.8

Smart_Stick_Continuous_Discharge_Raw.csv

elapsed_time_min	timestamp	battery_voltage_V	current_A	device_state
0	2026-09-01T08:00:00Z	4.2	0.42	active
30	2026-09-01T08:30:00Z	4.05	0.41	active
60	2026-09-01T09:00:00Z	3.91	0.4	active
90	2026-09-01T09:30:00Z	3.78	0.4	active
120	2026-09-01T10:00:00Z	3.65	0.39	active
150	2026-09-01T10:30:00Z	3.52	0.39	active
180	2026-09-01T11:00:00Z	3.38	0.38	active
210	2026-09-01T11:30:00Z	3.2	0.37	active
240	2026-09-01T12:00:00Z	3.01	0.35	cutoff_reached

Fusion_Weight_Sensitivity_Summary.csv

configuration	perturbation	all_10_baseline_tier_decisions_preserved	changed_decisions	notes
baseline	baseline	1	0	Reference configuration; unchanged by definition
C +20%	C +20%	0	1	One decision changed; same near-threshold case
C -20%	C -20%	1	0	
P +20%	P +20%	1	0	
P -20%	P -20%	0	1	One decision changed; same near-threshold case
U +20%	U +20%	1	0	

U -20%	U -20%	1	0	
W +20%	W +20%	1	0	
W -20%	W -20%	1	0	

TABLE S27. Material-dependent reflection and detectable-range characterization.

sensor_id	material	absorption	true_cm	measured_cm	rel_amp	max_range_cm	trial	valid	timestamp
U1	flat_board	0.12	100	100.3	1.00	350	1	1	2026-09-01T08:00:00Z
U1	flat_board	0.12	200	200.4	0.51	350	1	1	2026-09-01T08:00:05Z
U1	flat_board	0.12	300	300.2	0.24	350	1	1	2026-09-01T08:00:10Z
U2	concrete	0.35	100	101.8	0.78	320	1	1	2026-09-01T08:01:00Z
U2	concrete	0.35	200	202.1	0.38	320	1	1	2026-09-01T08:01:05Z
U2	concrete	0.35	300	303.5	0.16	320	1	1	2026-09-01T08:01:10Z
U3	metal	0.05	100	99.9	1.15	400	1	1	2026-09-01T08:02:00Z
U3	metal	0.05	200	199.8	0.62	400	1	1	2026-09-01T08:

									02:05Z
U3	metal	0.05	300	299.4	0.31	400	1	1	2026-09-01T08:02:10Z
U4	fabric	0.85	100	104.5	0.22	180	1	1	2026-09-01T08:03:00Z
U4	fabric	0.85	150	158.2	0.09	180	1	1	2026-09-01T08:03:05Z
U4	fabric	0.85	200	0.0	0.02	180	1	0	2026-09-01T08:03:10Z
U5	flat_board	0.12	100	100.1	1.01	350	2	1	2026-09-01T08:04:00Z
U5	flat_board	0.12	200	200.2	0.52	350	2	1	2026-09-01T08:04:05Z
U5	flat_board	0.12	300	299.9	0.25	350	2	1	2026-09-01T08:04:10Z
U1	concrete	0.35	100	101.6	0.79	320	2	1	2026-09-01T08:05:00Z
U1	concrete	0.35	200	201.9	0.39	320	2	1	2026-09-01T08:05:05Z

U1	concrete	0.35	300	303.1	0.17	320	2	1	2026-09-01T08:05:10Z
U2	metal	0.05	100	99.8	1.16	400	2	1	2026-09-01T08:06:00Z
U2	metal	0.05	200	199.6	0.63	400	2	1	2026-09-01T08:06:05Z
U2	metal	0.05	300	299.2	0.32	400	2	1	2026-09-01T08:06:10Z

TABLE S28. Temperature/humidity compensation records.

timest amp	sensor _id	temp_ C	humid ity_pct	sound_ speed_ m_s	true_c m	raw_c m	compe nsated _cm	error_c m	trial
2026-09-01T09:00:00Z	U1	20.0	45.0	343.46	100	101.2	100.0	0.0	1
2026-09-01T09:10:00Z	U1	22.5	47.0	345.00	100	101.8	100.1	0.1	1
2026-09-01T09:20:00Z	U1	25.0	50.0	346.50	100	102.3	100.1	0.1	1
2026-09-01T09:30:00Z	U1	27.5	52.0	348.01	100	102.9	100.0	0.0	1
2026-	U1	30.0	55.0	349.50	100	103.5	100.0	0.0	1

09- 01T09: 40:00Z									
2026- 09- 01T09: 50:00Z	U1	32.5	58.0	350.99	100	104.1	100.2	0.2	1
2026- 09- 01T10: 00:00Z	U1	35.0	60.0	352.48	100	104.7	100.2	0.2	1
2026- 09- 01T10:1 0:00Z	U1	37.5	62.0	353.96	100	105.4	100.1	0.1	1
2026- 09- 01T10: 20:00Z	U1	40.0	65.0	355.44	100	106.1	100.1	0.1	1
2026- 09- 01T10: 30:00Z	U1	42.0	68.0	356.64	100	106.7	100.0	0.0	1
2026- 09- 01T10: 40:00Z	U2	20.0	45.0	343.46	200	202.4	200.1	0.1	2
2026- 09- 01T10: 50:00Z	U2	25.0	50.0	346.50	200	204.6	200.2	0.2	2
2026- 09- 01T11:0 0:00Z	U2	30.0	55.0	349.50	200	207.1	200.0	0.0	2
2026- 09- 01T11:1	U2	35.0	60.0	352.48	200	209.5	200.3	0.3	2

0:00Z

2026-09-01T11:20:00Z	U2	40.0	65.0	355.44	200	212.2	200.2	0.2	2
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